

Problem

The voltage across an inductor leads the current through it by 90° .

- (a) True
(b) False

Step-by-step solution

Step 1 of 3

Assume that the expression for inductor current is,

$$i_L = I_m \sin \omega t \text{ A}$$

The phasor representation of inductor current is,

$$\mathbf{I}_L = I_m \angle 0^\circ \text{ A} \dots\dots (1)$$

Determine the voltage across the inductor using following formula.

$$\begin{aligned} v_L &= L \frac{di_L}{dt} \\ &= L \frac{d}{dt} (I_m \sin \omega t) \\ &= LI_m \frac{d}{dt} (\sin \omega t) \\ &= LI_m \cos \omega t \text{ V} \end{aligned}$$

Step 2 of 3

Apply the trigonometry property.

$$\sin(\omega t + 90^\circ) = \cos \omega t$$

Write the inductor voltage expression using this property.

$$\begin{aligned} v_L &= LI_m \cos \omega t \\ v_L &= LI_m \sin(\omega t + 90^\circ) \end{aligned}$$

The phasor representation of inductor voltage is,

$$\mathbf{V}_L = LI_m \angle 90^\circ \text{ V} \dots\dots (2)$$

Step 3 of 3

From equations (1) and (2), it is clear that the inductor voltage, $\mathbf{V}_L$ is leading the inductor current, $\mathbf{I}_L$ by 90° .

Therefore, the given statement is **True**.

Thus, the correct option is **(a)**.